Page and Block-wise Program Disturb Operation in NAND Flash Memory

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Introduction

The NAND flash memories are the heart of the data storage solution of commercial electronics. The memory cells are connected in an array (page) to store bits also referred to as program or erase state of the devices. However, the program or erase state of a specific page can be disturbed while other pages go through the programming cycle, which is also called program-disturb and is a key reliability issue in 2D NAND Flash memory (Fig. 1(a)) [1]. Here, we demonstrate different programming schemes: (i) page-wise program-disturb operation and (ii) block-wise program-disturb operation. Results highlight the impact on data due to page-wise and block-wise program disturb operation, including optimized programming algorithms to enhance the reliability and performance of 2D NAND Flash memory systems.

Experimental Details

The experimental setup includes a Xilinx Zybo Z7010 board with a TSOP package connected to a Micron memory chip (MT29F64G08CBABAWP: B) as shown in Fig. 1(b). The code for interfacing with the memory chip and controlling its read, write, and erase operations was written using the Software Development Kit (SDK) in Vivado [2]. In the page program-disturb experiment, data is written to page 1 across 80 blocks and then read back incrementally. This process is repeated for pages 2 and 3, with subsequent reads of page 1. The resulting data, shown in Fig. 3(a), plots the Fail Bit Count (FBC) [3] versus the read count of page 1, illustrating how writing to other pages affects previously stored bits on page 1 as illustrated in Fig. 3(a-inset). In the block program-disturb experiment, data is written to a single block across different pages, followed by reading the block. After programming the next block, block 1 is re-read. This process repeats with subsequent blocks, enabling analysis of program disturb effects. The algorithm is shown in Fig. 2(b) and Fig. 3(b-inset), with results discussed in the following section.

Results and Discussion

As shown in Fig. 3(c), page-wise programming (first method) results in a higher Fail Bit Count (FBC) compared to block-wise programming (second method). The FBC versus the number of reads for page-wise programming follows a Boltzmann fit with the coefficient of determination (R_2) equal to 0.99, highlighting three regions: an initial plateau with low disturb effects, a steep transition marking significant FBC growth, and a final plateau where FBC saturates as most cells are affected. Key parameters, such as the maximum FBC (A_1), minimum FBC (A_2), the transition point (x_0), and the slope (dx), provide insights into program disturb dynamics. The transition point (x_0) indicates the number of reads when disturb effects become prominent, while the slope (dx) reflects the rate of FBC increase. Boltzmann fitting thus aids in understanding program disturb behaviour and supports optimization strategies, such as block-wise programming, to enhance memory reliability. Here is the Boltzmann equation: $y = A_2 + \frac{A_1 - A_2}{1 + e^{\frac{x - x_0}{dx}}}$

The Fail Bit Count (FBC) versus the number of reads shows a linear trend in block-wise programming, indicating minimal program disturb compared to page-wise programming. This makes block-wise programming the preferred method to reduce data corruption. Fig. 3(a) illustrates that other pages within the same block are affected by program disturb in page-wise programming, while Fig. 3(b) confirms that block-wise programming results in lower FBC and reduced disturb. These findings highlight how the NAND Flash chip's internal structure and write methods influence program disturb behaviour.

Conclusion:

This study shows block-wise programming effectively reduces program disturb compared to page-wise programming in 2D NAND Flash memory. Boltzmann fitting of Fail Bit Count (FBC) vs. reads provides key insights. The transition point x0, representing the number of read threshold steps where disturb effects become significant, is observed at **[4.38 ~ 5 reads]**, slope dx, and FBC range [A_1-A_2], highlighting disturb severity. Understanding these parameters enables optimized memory operations, improving performance and longevity.

References

[1] Lou, Bojie, et al. "A Review of Program disturb of 3D NAND Flash Memory." *2023 24th International Conference on Electronic Packaging Technology (ICEPT)*. IEEE, 2023. [2] *AMD*. https://www.xilinx.com/support/download.html, [3] Mielke, Neal, et al. "Bit error rate in NAND flash memories." *2008 IEEE International Reliability Physics Symposium*. IEEE, 2008.

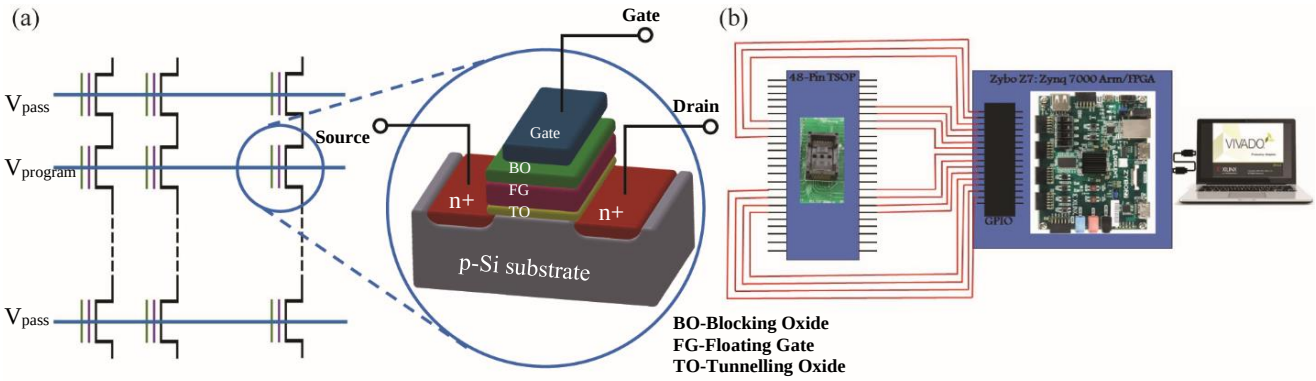

Fig. 1: (a) Schematic diagram of NAND structure with a 2D NAND cell showing programming a page by giving $V_{program}$ to program the page and V_{pass} to other pages (inset: structure of a 2D-NAND Flash cell) (b) Measurement setup for interfacing NAND Flash memory (MT29F6464G08CBABAWP: B) , using Xilinx Zybo Z7 FPGA , TSOP socket and Vivado SDK platform.

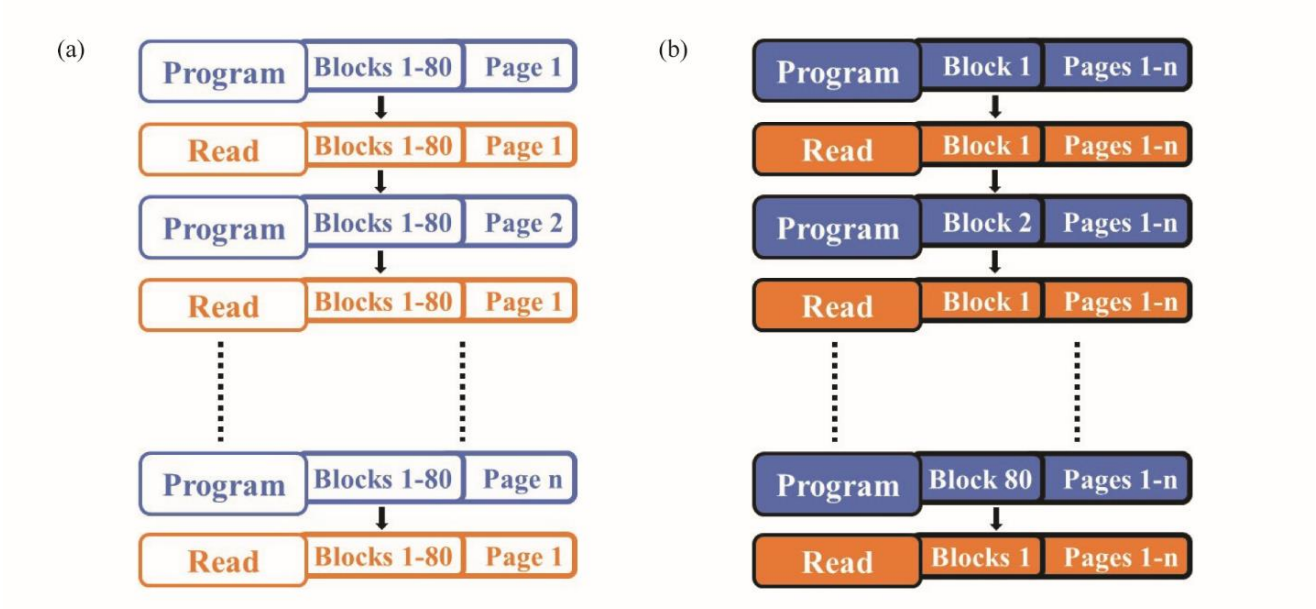

Fig. 2 : Algorithm : (a) Program first page of all the 80 blocks and reading first page, programming other pages and re-read the page 1 to check program disturb, (b) Same process but in terms of blocks.

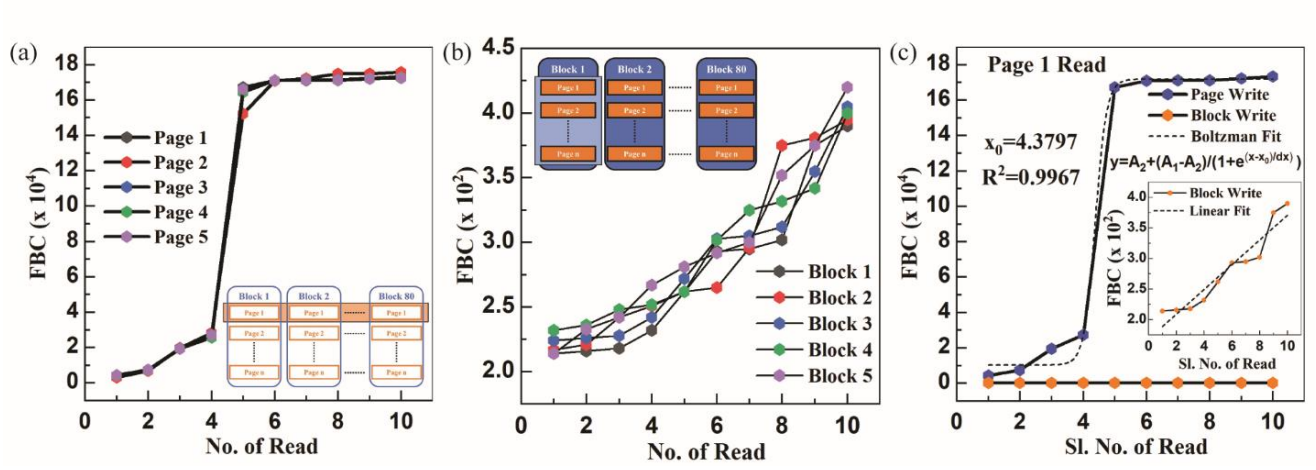

Fig. 3 : (a) FBC vs number of read of other pages by first method (inset - NAND Flash plane hierarchal sub-division of Blocks and Pages, Programming first page of all the 80 Blocks), (b) FBC vs number of read of other blocks by second method (inset - Programming all the pages of first block), (c) Comparison of FBC (Fail bit count) of both methods vs number of read of first page for first method with a fitting parameter and block for second method (inset – Linear Fitting parameter of second method).